

RHEUMATOID ARTHRITIS

Rheumatoid arthritis develops when the immune system produces antibodies that attack its own body tissues – especially the synovial membranes inside joints. The joints become swollen and deformed, with painful and restricted movement. Early general symptoms include fever, fatigue, and weakness. Characteristically, many of the small joints are affected in a symmetrical pattern; for example, the hands and feet may become inflamed to the same degree on both sides. Stiffness is often worse in the mornings but eases during the day. Painless small lumps or nodules (clusters of inflamed tissue cells), may form in areas of pressure, commonly on

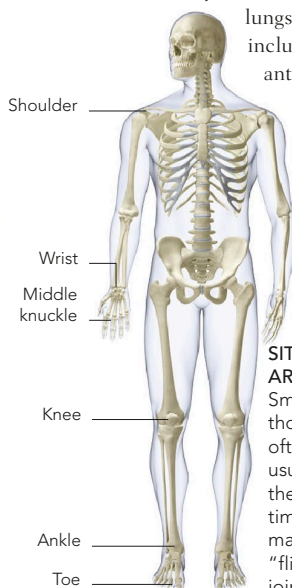


JOINT INFLAMMATION

In this X-ray, the middle knuckles of the hands are severely damaged by rheumatoid arthritis (red). Inflammation of the joints causes abnormal bending of the fingers.

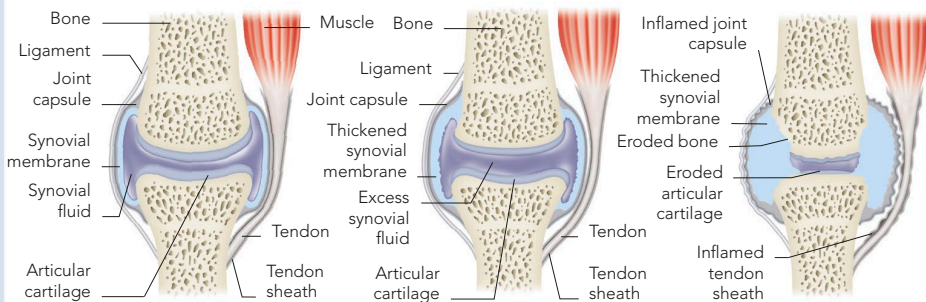
the forearms, and the skin over the joint is thin and fragile. The condition may flare up, then fade for a time. The diagnosis is supported if a blood test detects certain antibodies – RF and anti-CCP – associated with rheumatoid arthritis. The disease can also affect the eyes, skin, heart, nerves, and

lungs. Treatment includes simple anti-inflammatory drugs and disease-modifying antirheumatic drugs (DMARDs).



SITES OF RHEUMATOID ARTHRITIS

Smaller joints, especially those in the hands, are often affected first, usually on both sides of the body at the same time. The inflammation may then transfer or “flit” to other, larger joints, such as the wrist.



HEALTHY JOINT

Cartilage is smooth and intact in a healthy joint. Ligaments aid stability, and tendons slide in sheaths as muscles pull on them.

EARLY RHEUMATOID ARTHRITIS

The synovial membrane becomes inflamed and thickens, spreading across the joint. Excess synovial fluid accumulates in the joint.

LATE RHEUMATOID ARTHRITIS

As the synovial membrane thickens, the cartilage and bone ends are eroded. The joint capsule and tendon sheath become inflamed.